

Class #25 - Row space of A

- Def: Given an $m \times n$ matrix A , the *row space of A* is the set of all linear combinations of the rows of A and is denoted by $\text{Row}A$.

Note: $\text{Row } A = \text{Col } A^T$, and is a subspace of $\mathbb{R}^n$.

Note: $\text{rank } A = \dim \text{Col}A = \dim \text{Row}A$.

- Suppose $A \sim U$. Is $\text{Col}A = \text{Col}U$?

- Theorem: If two matrices A and B are row equivalent, then their row spaces are the same. If B is in echelon form, the non-zero rows of B form a basis for the row space of A as well as B .

(Note: This means $\text{Row}A = \text{Row}B$)

Proof:

$$4. \text{ Example: Let } A = \begin{bmatrix} -2 & -5 & 8 & 0 & -17 \\ 1 & 3 & -5 & 1 & 5 \\ 3 & 11 & -19 & 7 & 1 \\ 1 & 7 & -13 & 5 & -3 \end{bmatrix} \sim B = \begin{bmatrix} 1 & 3 & -5 & 1 & 5 \\ 0 & 1 & -2 & 2 & -7 \\ 0 & 0 & 0 & -4 & 20 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Note: The first 3 rows of B are linearly independent, does this mean the first 3 rows of A are linearly independent?

Note: Row operations do preserve the linear dependence relations among the *rows* of a matrix.

5. Example: A scientist has found two solutions to a homogeneous system of 40 equations in 42 variables. The two solutions are not multiples, and all other solutions can be constructed by adding together appropriate multiples of these two solutions. Can the scientist be certain that an associated nonhomogeneous system (with the same coefficients) has a solution?

6. The Invertible Matrix Theorem (continued):

Let A be an $n \times n$ matrix. Then the following are all equivalent to the statement that A is invertible.

- (m) The columns of A form a _____ of $\mathbf{R}^n$.
- (n) $\text{Col} A = \underline{\hspace{2cm}}$
- (o) $\dim \text{Col } A = \underline{\hspace{2cm}}$
- (p) $\text{rank } A = \underline{\hspace{2cm}}$
- (q) $\text{Nul } A = \underline{\hspace{2cm}}$
- (r) $\text{nullity } A = \dim \text{Nul } A = \underline{\hspace{2cm}}$